SSP

Computer & Communication Department

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Modern Physics
Sheet 4
Solutions

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1. A TV tube operates with a 20 kV accelerating potential. (a) What is the maximum energy and momentum of X-rays produced from the TV set? (b) What is the minimum wavelength of these photons?

Solution

a)

$$E_{max} = eV = 20\ keV = 20 \times 10^{3} \times 1.6 \times 10^{-19} = 3.2 \times 10^{-15} J$$

$$P_{max} = \frac{E_{max}}{c} = \frac{3.2 \times 10^{-15}}{3 \times 10^{8}} = 1.1 \times 10^{-23}\ kg.m/s$$

b)

$$\lambda_{min} = \frac{hc}{K_{max}} = \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{3.2 \times 10^{-15}} = 6.2 \times 10^{-11}\ m = 0.062\ nm$$

2. X-rays of wavelength λ = 0.200 nm are aimed at a block of carbon. The scattered x-rays are observed at an angle of 45.0^0 to the incident beam. Calculate the wavelength of the scattered x-rays at this angle.

Solution

$$\lambda' = \lambda + \frac{h(1-\cos(\theta))}{mc} = 0.200 \times 10^{-9} + \frac{6.63 \times 10^{-34}(1-\cos(45))}{9.11 \times 10^{-31} \times 3 \times 10^{8}} = 0.201\ nm$$

3. X-rays with an energy of 300 keV undergo Compton scattering from a target. If the scattered rays are detected at 30^{o} relative to the incident rays, find (a) the Compton shift at this angle, (b) the energy of the scattered x-ray, and (c) the energy of the recoiling electron.

Solution

a)

$$\Delta\lambda = \frac{h(1-\cos(\theta))}{mc} = 3.25 \times 10^{-13}\ m$$

b)

$$\lambda = \frac{hc}{E} = \frac{hc}{300 \times 10^{3} \times 1.6 \times 10^{-19}} = 4.14 \times 10^{-12}\ m$$

$$\lambda' = \lambda + \Delta\lambda = 4.46 \times 10^{-12}\ m$$

$$E' = \frac{hc}{\lambda'} = 278\ keV$$

c)

$$K_e = E - E' = 300 - 278 = 22\ keV$$

4. Gamma rays (high-energy photons) of energy 1.02 MeV are scattered from electrons that are initially at rest. If the scattering is symmetric, that is, if θ = φ in Figure 3.24, find (a) the scattering angle θ and (b) the energy of the scattered photons.

Solution

Applying the conservation of momentum

y-component:

$$p' \sin(\theta) = p_e \sin(\varphi)$$

Since $\theta = \varphi$

$$p' = p_e$$

x-component:

$$p = p' \cos(\theta) + p_e \cos(\varphi)$$

$$p = 2p' \cos(\theta)$$

Multiply both sides by c

$$pc = 2p'c \cos(\theta)$$

$$E = 2E' \cos(\theta) \qquad (1)$$

In terms of energy, the equation for Compton shift is written as:

$$\frac{1}{E'} - \frac{1}{E} = \frac{1 - \cos(\theta)}{mc^2} \qquad (2)$$

Substituting E = 1.02 MeV and Solving 1 and 2 for E' and $\cos(\theta)$

$E' = 0.679$ MeV

$\theta = 41.3^0$

5. A photon of initial energy 0.1 MeV undergoes Compton scattering at an angle of 60^0. Find (a) the energy of the scattered photon, (b) the recoil kinetic energy of the electron, and (c) the recoil angle of the electron.

Solution

a)

$$\Delta\lambda = \frac{h(1-\cos(\theta))}{mc} = 1.22 \times 10^{-12}\, m$$

$$\lambda = \frac{hc}{E} = \frac{hc}{300 \times 10^3 \times 1.6 \times 10^{-19}} = 1.24 \times 10^{-11}\, m$$

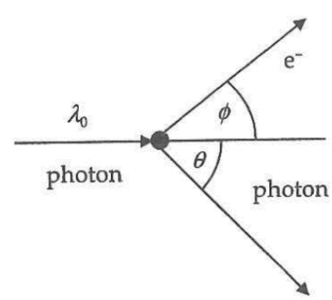
$$\lambda' = \lambda + \Delta\lambda = 1.36 \times 10^{-11}\, m$$

$$E' = \frac{hc}{\lambda'} = 91\, keV$$

b)

$$K_e = E - E' = 100 - 91 = 9\, keV$$

c)



$$p = \frac{E}{c} = 0.1\, MeV/c$$

$$p' = \frac{E'}{c} = 0.091\, MeV/c$$

$$p = p'\cos(\theta) + p_e\cos(\varphi)$$

$$p'\sin(\theta) = p_e\sin(\varphi)$$

Dividing both equations to eliminate p_e

$$tan(\varphi) = \frac{p' sin\,(\theta)}{p - p'\,cos(\theta)} = \frac{0.091\, sin\,(60)}{0.1 - 0.091\, cos\,(60)} = 1.446$$

$$\varphi = 55.3°$$

6. A photon undergoing Compton scattering has an energy after scattering of 80 keV, and the electron recoils with an energy of 25 keV. (a) Find the wavelength of the incident photon. (b) Find the angle at which the photon is scattered. (c) Find the angle at which the electron recoils.

Solution

a)

$E = E' + K_e = 80 + 25 = 105\ keV$

$\lambda = \frac{hc}{E} = 0.0118\ nm$

b)

$$\frac{1}{E'} - \frac{1}{E} = \frac{1 - \cos(\theta)}{mc^2}$$

$$\frac{1}{80} - \frac{1}{105} = \frac{1 - \cos(\theta)}{511}$$

$\theta = 121°$

c)

$$tan(\varphi) = \frac{p' sin\ (\theta)}{p - p'\ cos(\theta)} = \frac{80\ sin\ (121)}{105 - 80\ cos\ (121)} = 0.469$$

$\varphi = 25.1°$

7. If the maximum energy given to an electron during Compton scattering is 30 keV, what is the wavelength of the incident photon?

Solution

$K_e = E - E'$

$30 = E - E'$ (1)

To maximize K we must minimize E' so the scattering angle must be 180^0.

$$\frac{1}{E'} - \frac{1}{E} = \frac{1 - \cos(\theta)}{mc^2}$$

$$\frac{1}{E'} - \frac{1}{E} = \frac{2}{511} \qquad (2)$$

Solving (1) and (2)

$E = 104\ keV$

$$\lambda = \frac{hc}{E} = 1.19 \times 10^{-11}\ m$$

8. Show that a photon cannot transfer all of its energy to a free electron.

Solution

If the photon gives all of its energy to a free electron, then

$E' = p' = 0$

$E = K_e$

$p = p_e$

$pc = p_e c$

$K_e = p_e c$ (1)

Generally

${E_e}^2 = (p_e c)^2 + (mc^2)^2$

$(K_e + mc^2)^2 = (p_e c)^2 + (mc^2)^2$

${K_e}^2 + (mc^2)^2 + 2K_e mc^2 = (p_e c)^2 + (mc^2)^2$

From (1)

$(p_e c)^2 + (mc^2)^2 + 2K_e mc^2 = (p_e c)^2 + (mc^2)^2$

$2K_e mc^2 = 0$

Or

$Emc^2 = 0$

Since neither E, m nor c is equal to zero then the first hypothesis is incorrect, and the photon can’t give all of its energy to a free electron.

9. Determine the threshold wavelength for pair production.

Solution

$$E_{min} = 2m_e c^2 = 1.02\ MeV$$

$$\lambda_{max} = \frac{hc}{E_{min}} = 1.22 \times 10^{-12}\ m$$

10. A photon of wavelength 0.003 A in the vicinity of a heavy nucleus produces an electron-positron pair. The kinetic energy of the positron is twice that of the electron. Determine the kinetic energy of both particles.

Solution

$$E = \frac{hc}{\lambda} = 4.14\ MeV$$

$$E = 2m_e c^2 + K_- + K_+$$

$$K_+ = 2K_-$$

$$E = 2m_e c^2 + 3K_-$$

$$K_- = \frac{4.14 - 2 \times 0.511}{3} = 1.04\ MeV$$

$$K_+ = 2K_- = 2.0\ MeV$$